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Global weak solutions of an initial boundary value problem for screw pinches in plasma physics

（箍缩）

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Outline：

1. Governing equations
2. Introduction of weak solutions
3. Global existence
4. Idea of the proof

1. Governing Equations

MHD concerns the motion of a conducting fluid (plasma) in an electromagnetic field with a very wide range of applications. The dynamic motions of the fluid and the magnetic field strongly interact each other, and thus both the hydrodynamic and electrodynamic effects have to be considered. The general governing equations of 3D MHD read:

1. Governing equations (General case)

$$\rho_t + \mathrm{div}\,(\rho u) = 0,$$

$$(\rho u)_t + \mathrm{div}\,(\rho u \otimes u) + \nabla p = (\nabla \times B) \times B + \mathrm{div}\,\mathbb{S},$$

$$B_t - \nabla \times (u \times B) = -\nabla \times (u \nabla \times B), \ \mathrm{div} B = 0,$$

$$\left(\frac{1}{2}\rho|u|^2 + \rho e + \frac{1}{2}|B|^2\right)_t + \mathrm{div}\,\left(\left(\frac{1}{2}\rho|u|^2 + \rho e + p\right)u - \kappa \nabla \theta\right) = \mathrm{div}\,((u \times B) \times B + \nu B \times (\nabla \times B) + \mathbb{S}u).$$

Here, ρ: Density, $u \in \mathbb{R}^3$: Velocity,
$B \in \mathbb{R}^3$: Magnetic field, θ: Temperature,

$p = R\rho\theta$: Pressure, $e = c_v\theta$: Internal energy, $\mathbb{S}$: Viscous stress tensor

$$\mathbb{S} = \lambda'(\mathrm{div}\, u)\mathbb{I} + \mu(\nabla u + (\nabla u)^\top),$$

λ', μ: Viscosity coefficients,
$\kappa = \kappa(\rho, \theta)$: Heat conductivity,
ν: Magnetic diffusivity

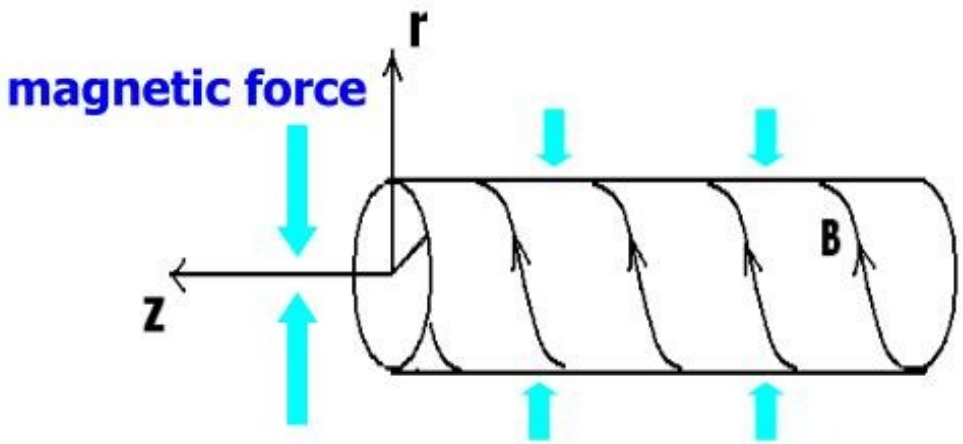
A complete system for ρ, u, θ, B !

The system is too complex to study mathematically.

Let's consider a special but physically very interesting case: Screw Pinches which have very important applications in plasma physics.

• Screw pinch case

Consider an cylindrical column of a (plasma) fluid with an axial current density and a resulting azimuthal magnetic induction. Thus, the magnetic force, acting on the plasma, forces the plasma column to constrict radially. This radial constriction is known as the pinch effect (first by Bennett '34)

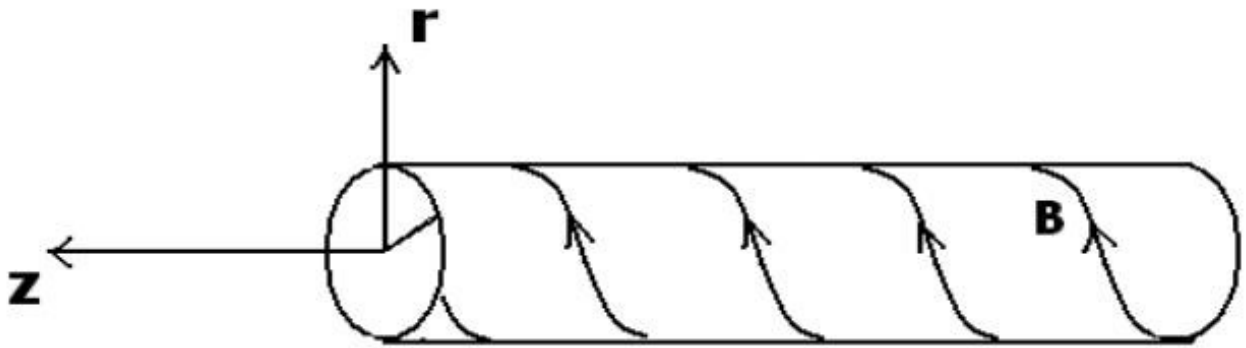


• Screw pinch case

Write u, B in cylindrical coordinates (r, ϕ, z) **:**

$$u = (u_r, u_\phi, u_z), \quad B = (B_r, B_\phi, B_z)$$

In the screw pinch, the magnetic field lines are sheared in the radial direction (i.e., the lines wind around the axis in a helical path), i.e., $B_r = 0$ **, while** $B_\phi = B_\phi(r,t), \quad B_z = B_z(r,t).$



Screw Pinch: Magnetic field lines wind around the axis in a helical path

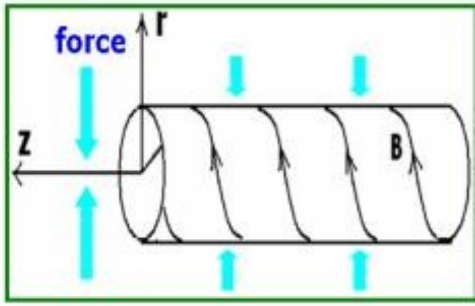
Consider the cylindrical plasma fluid without swirl, then

$$\rho = \rho(r,t),\ \theta = \theta(r,t),$$
$$u_r = u_r(r,t),\ u_\phi = u_z = 0.$$

Remark: **Screw pinch -- 1) In (x,y,z) coordinates:**

$$B = (-B_\phi \sin\phi, B_\phi \cos\phi, B_z)^\top,\ u = (u_r \cos\phi, u_r \sin\phi, 0)^\top$$

2) In (r,ϕ,z) coordinates: $B = (0, B_\phi, B_z)$, while the current density $J = (0, J_\phi, J_z) \Rightarrow$ the force $F = J \times B$: directed toward to the z-axis, constricting "hot" plasma to the axis.



Hence, in the screw pinch case the general MHD equations becomes the following system for

$(\rho, u_r, B_\phi, B_z, \theta)$, denoted by $(\rho, u, B_1, B_2, \theta)$ for simplicity: $(x \equiv r)$

$$\rho_t + (\rho u)_x + \frac{\rho u}{x} = 0, \tag{1}$$

$$\rho(u_t + u u_x) + p_x + \frac{(B_1^2 + B_2^2)_x}{2} + \frac{B_1^2}{x}$$
$$= \lambda(u_x + \frac{u}{x})_x, \tag{2}$$

$$\partial_t B_1 + (u B_1)_x = \nu(B_{1x} + \frac{B_1}{x})_x, \tag{3}$$

$$\partial_t B_2 + (u B_2)_x + \frac{u B_2}{x} = \nu(B_{2xx} + \frac{B_{2x}}{x}), \tag{4}$$

$$c_v \rho(\theta_t + u\theta_x) + p(u_x + \frac{u}{x})$$
$$= (\kappa\theta_x)_x + \frac{\kappa\theta_x}{x} + \mathcal{Q}, \tag{5}$$

where $p = R\rho\theta, \quad (\lambda = \lambda' + 2\mu)$

$$\mathcal{Q} = \lambda'(u_x + \frac{u}{x})^2 + 2\mu\left[u_x^2 + (\frac{u}{x})^2\right]$$
$$+\nu\left[(B_{1x} + \frac{B_1}{x})^2 + B_{2x}^2\right],$$

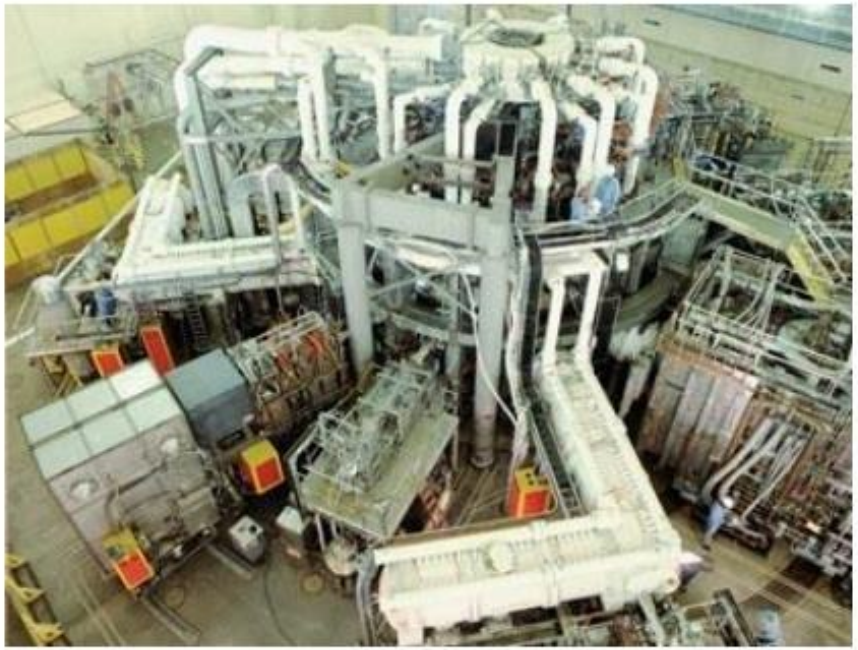
with initial and boundary conditions:

$$(\rho, u, B_1, B_2, \theta)|_{t=0}$$
$$= (\rho_0, u_0, B_{10}, B_{20}, \theta_0), \ x \in (0,1), \quad (6)$$
$$(u, B_1, B_2, \theta_x)|_{x=1} = 0, \ t \geq 0. \quad (7)$$

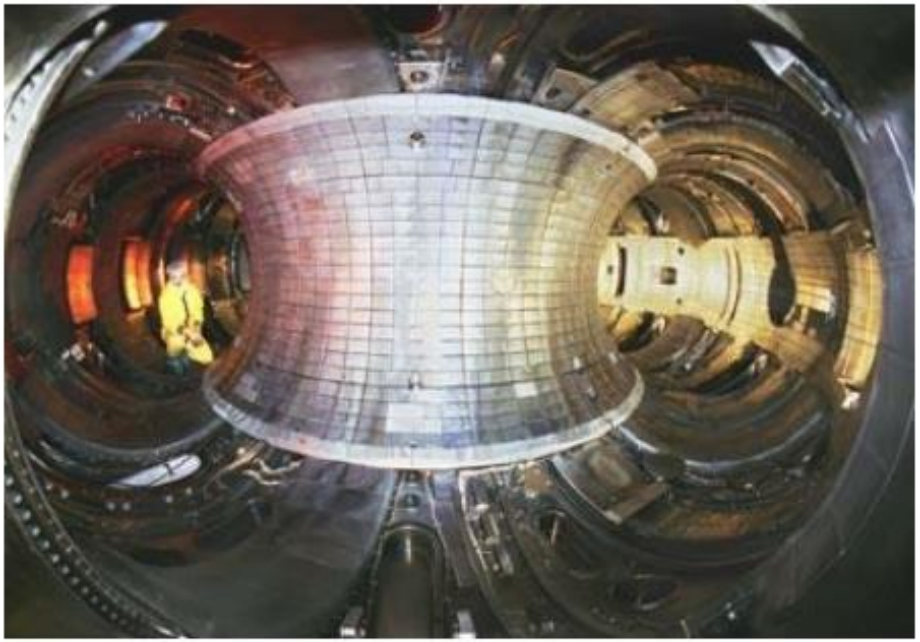
Remarks:

1) Screw pinches have important applications in physics of plasmas, e.g., "Tokamak" devises which confine and constrict "hot" plasma to realize nuclear fusion in labs.

Princeton Plasma Physics Laboratory



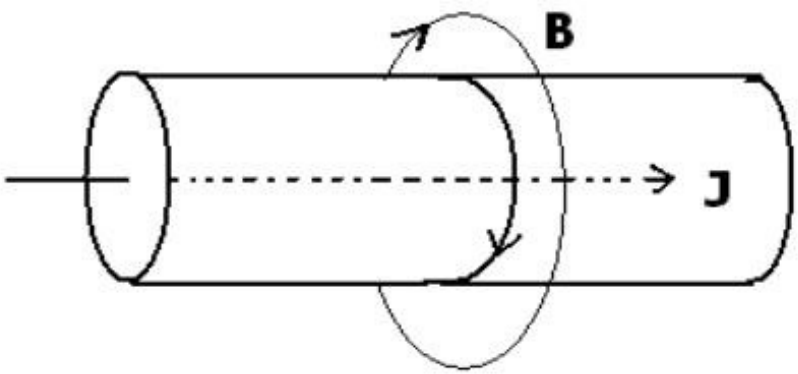
Tokamak Fusion Test Reactor (TFTR)



Inside the TFTR Vacuum Vessel

2) When $B = (0, B_\phi, 0)$ ($J = (0, 0, J_z)$), it is called Z-pinch.
When $B = (0, 0, B_z)$ ($J = (0, J_\phi, 0)$), it is called ϕ-pinch $\Rightarrow$

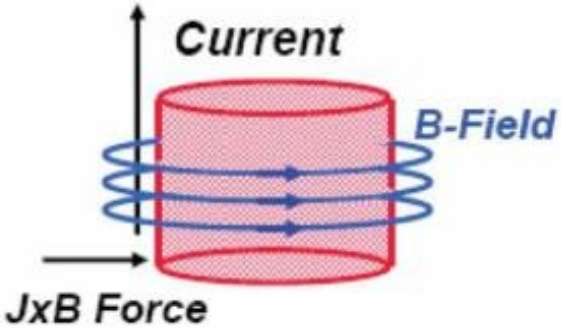
screw pinch = Z-pinch + ϕ-pinch !

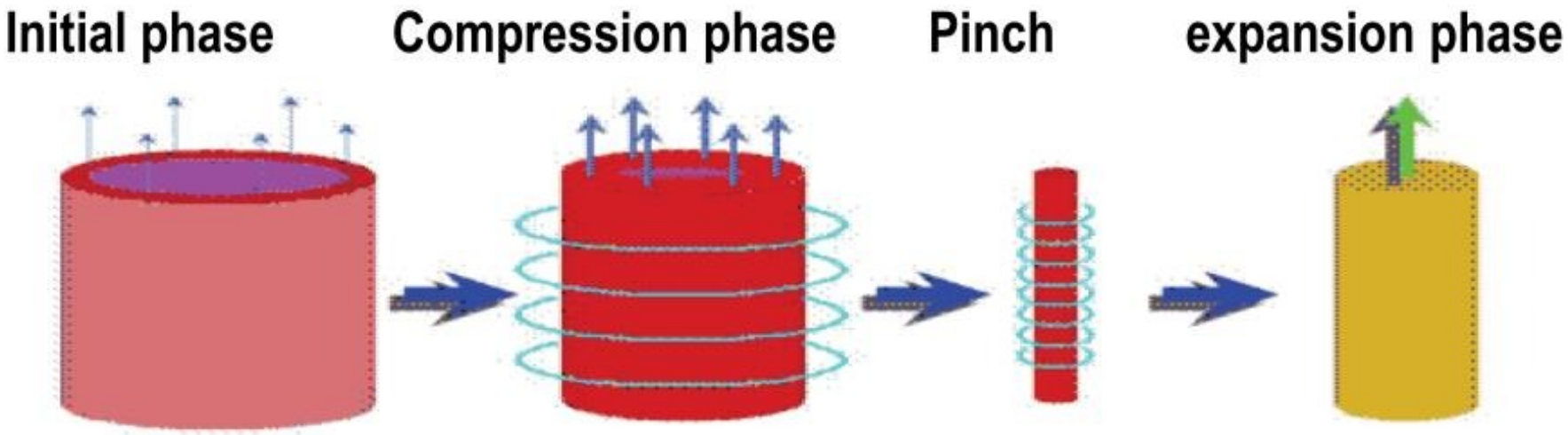


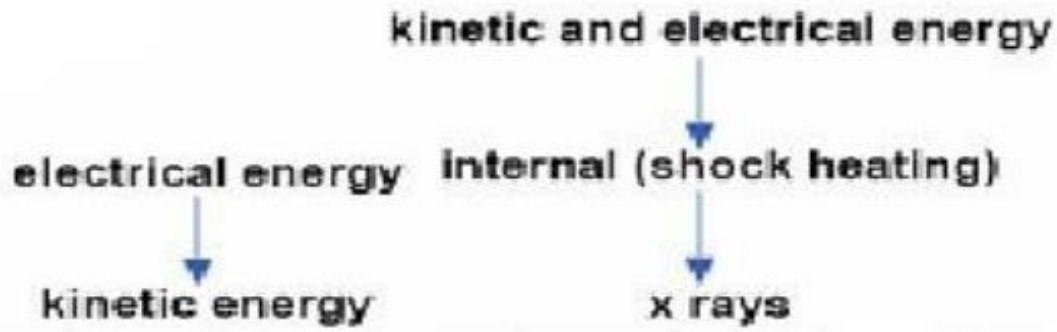
Z-Pinch

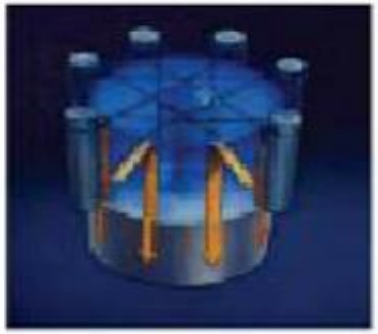
Z-pinch devise is another important possibility to realize nuclear fusion in labs.

JxB: directed toward to the z-axis



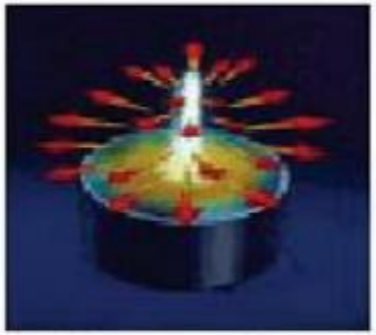






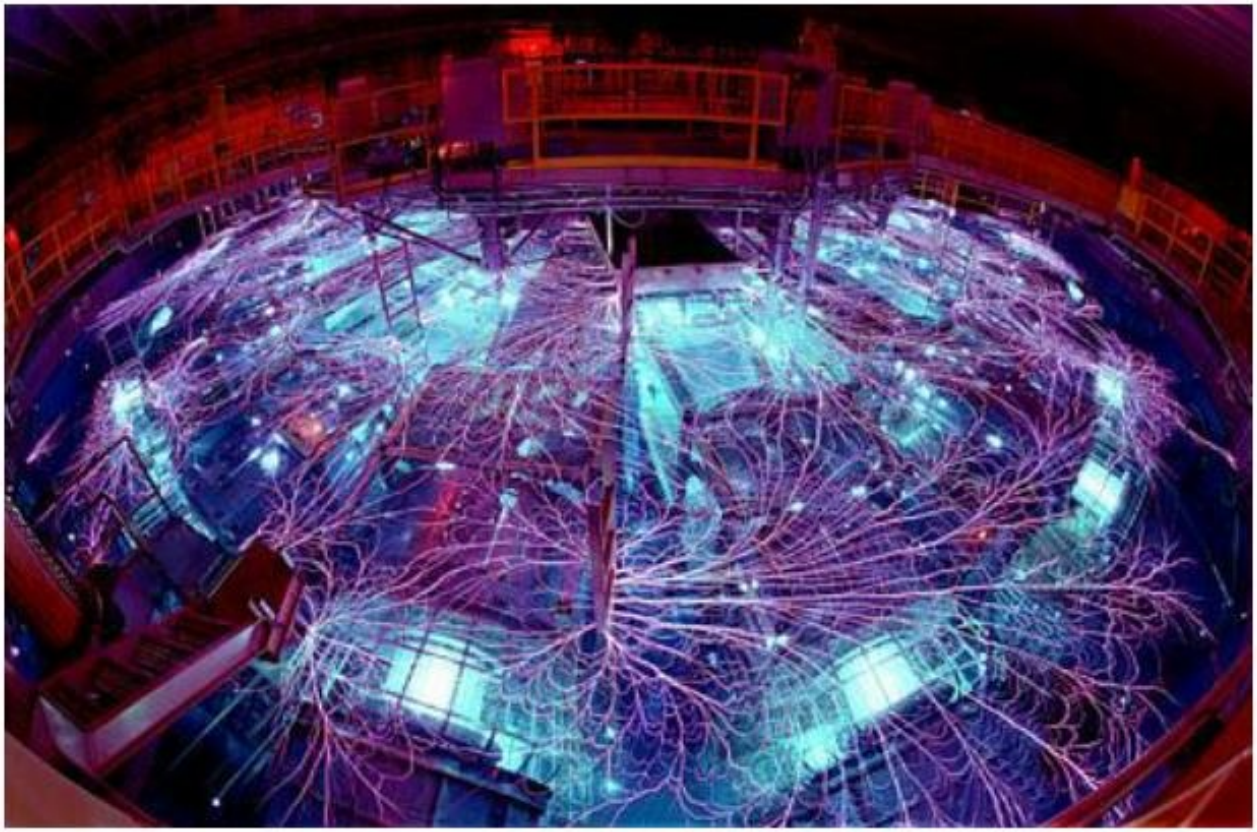
Initiation

Implosion



Stagnation

Sandia's Z-Accelarator



Time-exposure photograph of electrical flashover arcs produced over the surface of the water in the accelerator tank as a byproduct of Z operation. These flashovers are much like strokes of lightning

Mathematical difficulties:

1. Singularity at x=0.
2. Strong coupling (interaction) of the magnetic field and fluids
3. Strong nonlinearities
4. Degenerate at $\rho = 0$ or ∞

Aim of this talk: ***To show the existence of a global "weak solution" to the screw pinch problem (1)—(7) !***

2. Definition of weak solutions

Actually, our global weak solution $(\rho, u, B_1, B_2, \theta)$ to (1)–(7) has the following properties:

For any $T > 0$,

a) $\rho,\ \theta \geq 0$ a.e.,

$$\rho, \rho\theta \in L^\infty(0,T;\mathcal{L}^1),\ \sqrt{\rho}u, B_1, B_2 \in L^\infty(0,T;\mathcal{L}^2),$$

$$\frac{\sqrt{\kappa}\theta_x}{\theta},\ \frac{u_x}{\sqrt{\theta}} \in L^2(0,T;\mathcal{L}^2),$$ (energy space)

where $\mathcal{L}^p = L^p(0,1)$ with the measure xdx.

Similarly, $\mathcal{H}^1 = H^1(0,1)$ with the measure xdx

b) supp(ρ) is bounded on the left by a semi-continuous curve $\underline{x}(t) : [0, \infty) \to [0, \infty)$. Moreover, if $\mathcal{F}$ is the "fluid region", defined by $\mathcal{F} := \{(x,t) : t \geq 0, \quad \underline{x}(t) < x \leq 1\}$, then $\mathcal{F} \cap \{t > 0\} \cap \{x < 1\}$ is an open set.

c) $(\rho, \theta) \in L^\infty_{\text{loc}}(\mathcal{F})$, $(\rho, u, B_1, B_2, \theta)$: locally Hölder continuous in $\mathcal{F} \cap \{t \geq 0\}$. Eqs. (1)–(5) hold in $\mathcal{D}'(\mathcal{F} \cap \{t \geq 0\} \cap \{x < 1\})$.

d) $(\rho, u, B_1, B_2, \theta)$ satisfies the weak form of Eqs.(1)-(4) on the whole domain $(0,1)\times(0,T)$.

e) For $\psi \in C^2([0,1] \times [t_1, t_2])$, $\operatorname{supp} \psi(\cdot, t) \subset \{x : x \geq \underline{x}(t) + \eta\}$ with $\eta > 0$ for each $t \in [t_1, t_2]$, the energy eq.(5) holds in the sense:

$$\int_0^1 c_v x \rho \theta \psi dx|_{t=t_1}^{t=t_2} - \int_{t_1}^{t_2} \int_0^1 c_v x \rho \theta (\psi_t + u \psi_x) dx dt$$
$$+ \int_{t_1}^{t_2} \int_0^1 \gamma x \rho \theta (u_x + \frac{u}{x}) \psi dx dt$$
$$= \int_{t_1}^{t_2} \int_0^1 x \mathcal{Q} \psi dx dt - \int_{t_1}^{t_2} \int_0^1 x \kappa(\rho, \theta) \theta_x \psi_x dx dt.$$

(The energy equation is satisfied in the no-vacuum region.)

f) The total energy

$$\mathcal{E}(t) := \int_0^1 x(c_v \rho\theta + \frac{\rho u^2}{2} + \frac{B_1^2 + B_2^2}{2})(x,t)dx$$

satisfies

$$\mathcal{E}(t) = \mathcal{E}(0) - \lim_{R\to 0}\lim_{j\to\infty}\int_{\epsilon_j}^{R} x(c_v \rho^j \theta^j + \frac{\rho^j (u^j)^2}{2})dx,$$

where $0 < \epsilon_j \to 0$ as $j \to \infty$; $(\rho^j, u^j, B_1^j, B_2^j, \theta^j)$ are the approximate smooth solutions in the annular domains $\Omega_j := \{x : \epsilon_j < x < 1\}$.

Then, the main result reads:

3. Global existence

Thm. 1) *Assume* $C^{-1} \le \rho_0, \theta_0 \le C$, *and*

$$\rho_0 \in \mathcal{H}^1, \quad B_{10}, B_{20}, \theta_0 \in \mathcal{L}^2,$$

$$\rho_0,\ \rho_0\theta_0,\ \rho_0 u_0^2,\ \ \rho_0 g(\rho_0^{-1}),\ \rho_0 g(\theta_0) \in \mathcal{L}^1$$

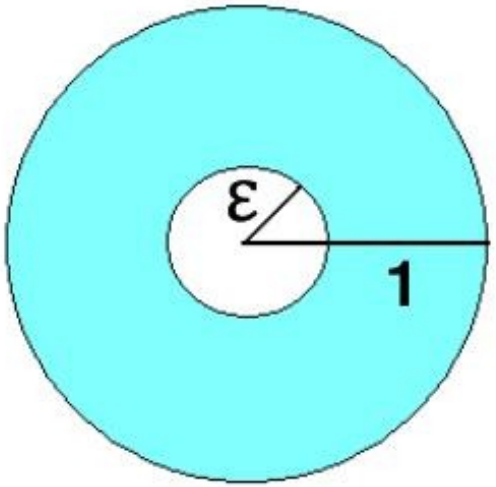
where $g(s) = s - \ln s - 1$. *Also assume*

$$\kappa_1(1+\theta)^q \le \kappa(\rho,\theta), \quad |\kappa_\rho(\rho,\theta)| \le \kappa_2(1+\theta)^q$$

for some $q \ge 2$.

Then, $\exists$ *a global weak solution* $(\rho, u, B_1, B_2, \theta)$ *of (1)–(7) on* $(0,1) \times (0,T)$, *satisfying the above a) – f).*

4. Proof steps



Proof idea: **Consider the problem in the annular $\{\epsilon < x < 1\}$ domains where no singularity is present in the equations, and obtain thus the approximate solutions.**

Then, with the help of the uniform in ε estimates in the energy space, we take the limit $\varepsilon \to 0$ for the approximate solutions to show that the limit is the desired weak solution.

Proof Steps:

i) Approximate solutions

Consider the problem (1)-(7) in the domain $\{\epsilon < x < 1\}$ **with additional boundary condition:**

$$((u, B_1, B_2, \theta_x)|_{x=\epsilon} = 0. \qquad (8)$$

Since no singularity in the equations, there exists a global strong solution $(\rho^\epsilon, u^\epsilon, B_1^\epsilon, B_2^\epsilon, \theta^\epsilon)$ **to (1)-(8).**

ii) (Global) Uniform in ε estimates estimate

standard energy estimates

$$\sup_{[0,T]} \int_\epsilon^1 x \left[\rho^\epsilon + \frac{\rho^\epsilon (u^\epsilon)^2 + (B_1^\epsilon)^2 + (B_2^\epsilon)^2}{2} + c_v \rho^\epsilon \theta^\epsilon \right] (x,t) dx \leq C,$$

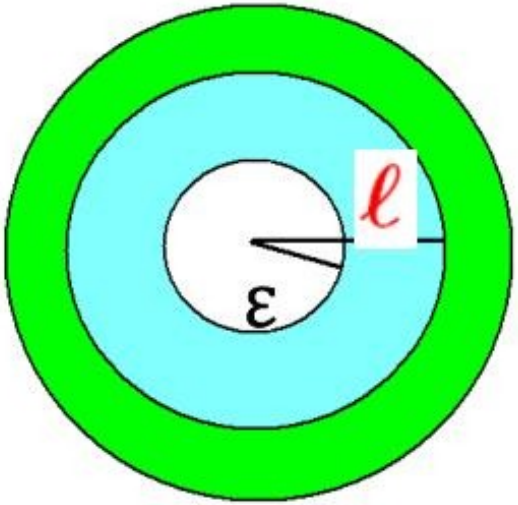
$$\sup_{[0,T]} \int_\epsilon^1 x\rho^\epsilon \left[g(1/\rho^\epsilon) + g(\theta^\epsilon)\right](x,t)dx$$
$$+ \int_0^T \int_\epsilon^1 x \left(\frac{\kappa(\theta_x^\epsilon)^2}{(\theta^\epsilon)^2} + \frac{\mathcal{Q}}{\theta^\epsilon} \right) dxdt \leq C.$$

These are all uniform <u>global</u> estimates we can have, with the help of which we have to pass to the limit as $\varepsilon \to 0$ and to show the global existence !

Next we want to show for any $\ell > \epsilon$, **(away from 0)**

$$c_\ell \leq \rho^\epsilon(x,t) \leq C_\ell, \quad \forall\, x \geq \ell,$$

c_ℓ, C_ℓ: constants independent of ϵ.



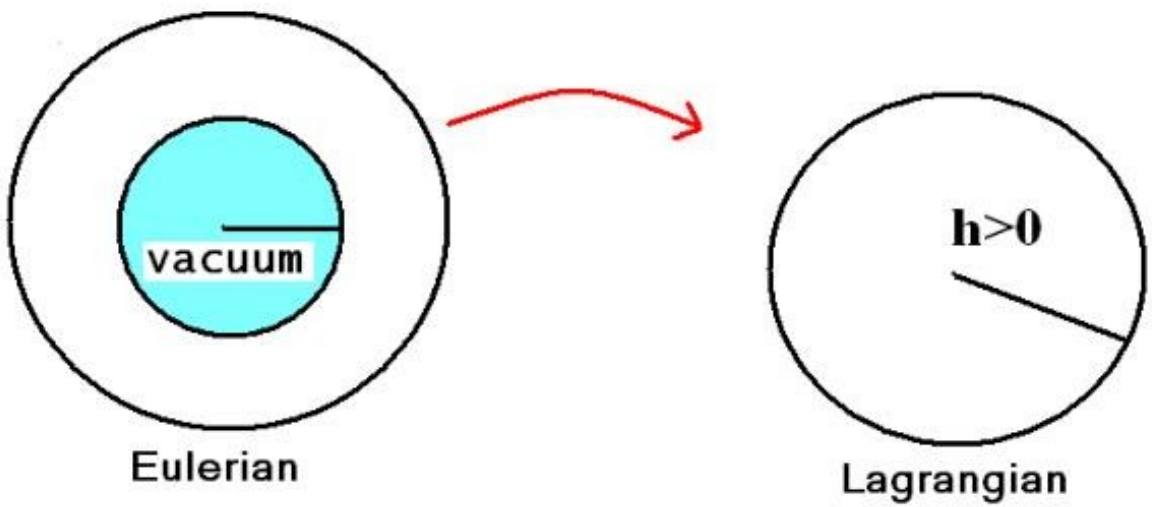
This can be shown only in Lagrangian coordinates,

● **Introduce the Lagrangian coordinates to get more estimates:** **For *h≥0*, define the curve** $x_h^\epsilon(t)$ by

$$h = \int_\epsilon^{x_h^\epsilon(t)} x\rho^\epsilon(x,t)dx \Rightarrow \frac{d}{dt}x_h^\epsilon(t) = u^\epsilon(x_h^\epsilon(t),t),$$

$$x_h^\epsilon(t) \geq C(h) > 0 \text{ if } h > 0.$$

★ $\rho^\epsilon > 0 \Rightarrow h > 0$, but $h > 0 \not\Rightarrow \rho^\epsilon > 0$!



h=0 → $x_h^\epsilon(t) > 0$!

● More uniform (local) estimates <u>away from</u> the origin *h=0* of Lagrangian space

$$\text{For } h>0, \quad C(h,T)^{-1} \le \rho^\epsilon(x,t) \le C(h,T) \qquad (9)$$

for any $(x,t) \in [x_h^\epsilon(t), 1] \times [0,T]$.

$$\int_0^T (\|\theta^\epsilon\|_{h,\infty}^q + \|(B_1^\epsilon, B_2^\epsilon)\|_{h,\infty}^2) dt \le C(h,T), \qquad (10)$$

where $\|\cdot\|_{h,\infty} := \|\cdot\|_{L^\infty(\Omega_h^\epsilon)}$, $\Omega_h^\epsilon := (x_h^\epsilon(t), 1)$.

Also, derivatives can be similarly bounded:

$$\rho_x^\epsilon, u_x^\epsilon, (B_1^\epsilon, B_2^\epsilon)_x, \theta_x^\epsilon, \rho_t^\epsilon \in L^\infty(0,T; \mathcal{L}^2(x_h^\epsilon(t),1)),$$

$$u_t^\epsilon, \partial_t B_1^\epsilon, \partial_t B_2^\epsilon, \theta_t^\epsilon \in L^2(0,T; \mathcal{L}^2(x_h^\epsilon(t),1)).$$

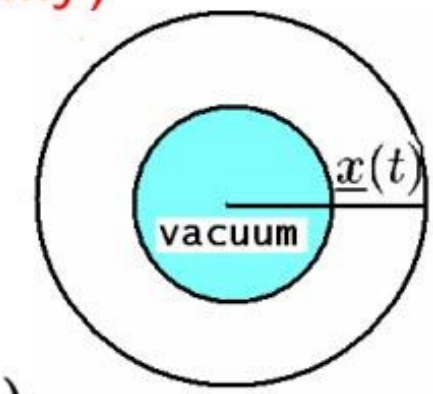
.....

iii) Limit process

Since $x_h^\epsilon(t)$ Holder continuous in (h, t) for h>0, t≥0 ⇒

$$\underline{x}(t) = \lim_{h\to 0}\lim_{\epsilon\to 0} x_h^\epsilon(t), \quad x_h(t) = \lim_{\epsilon\to 0} x_h^\epsilon(t) \text{ exists.}$$

Moreover, $\lim_{t\to 0} \underline{x}(t) = 0$ (since no vacuum initially)



Define *"fluid region"*

$$\mathcal{F} := \{(x,t) \mid \underline{x}(t) < x \le 1,\ t \ge 0\}.$$

($(0,1)\times(0,T)\backslash\mathcal{F}$: vacuum region)

Uniform bounds of derivatives away from the Lagrangian origin $(h > 0)$ ⇒

$$(\rho^\epsilon, u^\epsilon, B_1^\epsilon, B_2^\epsilon, \theta^\epsilon) \to (\rho, u, B_1, B_2)$$

uniformly in $K \subset\subset \mathcal{F}$ (away from $h = 0$)

Hence, the limit functions satisfy the problem (1)–(7) in $\mathcal{F}$ in the weak sense $\Rightarrow$ singularity in the weak convergence could only occur in the vicinity of $x = 0$!

To exclude possible concentration of singularity at $x = 0$, we need the following lemma:

Lemma 2. *Let $f : [\epsilon, 1] \to \mathbb{R}$ be non-negative and satisfy $|\int_\epsilon^1 x f \log f dx| \leq C_1$. Then, $\exists\, \omega(r, s)$ with $\lim_{r \to 0} \omega(r, s) = 0$, such that for any measurable $E \subset [\epsilon, 1]$,*

$$\int_E x f dx \leq \omega(|E|, C_1), \quad |E| := \int_E x dx.$$

Recalling

$$\int_{\epsilon}^{1} x\rho^{\epsilon} \log \rho^{\epsilon} dx \leq C,$$

use Lemma 2, the global uniform estimates in Step ii) and delicate analysis $\Longrightarrow$

No concentration of singularity around $x = 0$ for eqs. (1)–(4) (mass & momentum equations); for example,

convergence of the term $I := x\rho^\epsilon u^\epsilon - x\rho u$ in the weak form of (1):

$$
\begin{aligned}
&\left|\int_{t_1}^{t_2}\int_0^1 I\phi_x\right| = \left|\int_{t_1}^{t_2}\left(\int_0^{x_h(t)} + \int_{x_h(t)}^1\right) I\phi_x\right| \\
&\le C\int\left(\int_0^{x_h^\epsilon(t)} + \int_{x_h^\epsilon(t)}^{x_h(t)}\right) x\rho^\epsilon|u^\epsilon| + C\iint_{\underline{x}(t)}^{x_h(t)} x\rho|u| \\
&\quad + \left|\iint_{x_h(t)}^1 I\phi_x\right| \\
&\le C\sqrt{h + \omega(|[x_h^\epsilon(t), x_h(t)]|, C)} \\
&\quad + C\iint_{\underline{x}(t)}^{x_h(t)} x\rho|u| + \left|\iint_{x_h(t)}^1 I\phi_x\right| \to 0 \qquad (\varepsilon \to 0,\ h \to 0)
\end{aligned}
$$

The rest terms in Eqs. (1)-(4) can be similarly treated !

However, we can not exclude the concentration for Eq. (5) (energy eq.), which is included in our " weak solution" ~ our weak solution is in the generalized weak sense !

□

Remarks:

i) Related results: For the 3D case, Ducomet & Fereisl '05 proved the existence of so-called "variational solutions under strong growth conditions on *p, e, et al.* However, the polytropic gas case studied here is excluded.

"Variational solution" ~ mass & momentum Eqs. hold in the weak sense, but the energy Eq. holds in the form of weak inequality and the energy inequality holds.

1D: **G.Q. Chen & D.H. Wang, weak and smooth solutions**

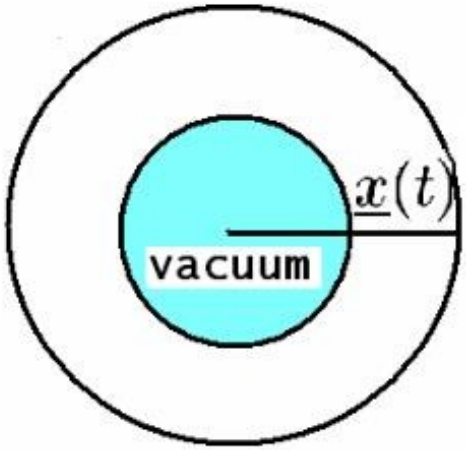
……

A similar result by Jenssen & Hoff for the compressible N-S equations '06, but they did not exclude singularity in the momentum eqns., i.e., the momentum eqs. do not hold in the classical sense of weak solutions.

ii) **Our growth condition on** $\kappa(\rho, \theta)$ **is physically valid for many physical regimes (high temperature):**

$\kappa \sim \theta^{5/2}$, or $\kappa \sim \theta^3$ **(equilibrium diffusion theory)**

iii) **We do not have sufficient information to determine whether** $\underline{x}(t) = 0$**. If** $\underline{x}(t) > 0$**, a vacuum state of radius** $\underline{x}(t)$ **centered at x=0 emerges. In both cases, the total mass is conserved.**



iv) **For our weak solution, the energy eq. (5) holds only on supp(*ρ*). This is mainly due to the possibility that vacuum states may arise, and we thus can not interpret the viscous terms and the term $\rho\theta(u_x + u/x)$ as distributions in the whole domain.**

It is also reasonable that the energy Eq. holds on supp(*ρ*), since no fluid outside $\mathcal{F}$, and the model is not valid there.

Thanks！

谢谢！

v) **Concerning the total energy** $\mathcal{E}$ **in** **f)** **in** **Definition of Weak Solutions, if our solution is smooth, then** $\mathcal{E}(t) = \mathcal{E}(0)$. **However, for our weak solution we have only** $\mathcal{E}(t) \leq \mathcal{E}(0)$ $\rightsquigarrow$ **the total energy could possibly absorbed into the origin.**